MOHAMED AMINE TRABELSI

Data & AI Engineering
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PROFILE SUMMARY

Data & AI Engineering student (Bac+5 / Master-level Cycle d'Ingénieur) with hands-on experience building full-stack applications, Retrieval-Augmented Generation (RAG) pipelines, and computer vision systems across four industry internships. Skilled at architecting provider-agnostic LLM integrations (Mistral, Gemini, MedGemma) and delivering production-ready APIs with FastAPI, Django, and Flask. Demonstrated expertise in designing end-to-end pipelines from data ingestion through deployment, backed by 5 industry-recognized DataCamp certifications. Seeking a challenging internship to apply this expertise to impactful, scalable AI-driven solutions.

PROFESSIONAL EXPERIENCE

06/2026 - 08/2026 | IdeaLab — Bizerte, Tunisia

AI / RAG Engineer (Mistral / Gemini) & Full-Stack Developer (Python / TypeScript)

- Design and development of Digital Campus end-to-end, a full-stack academic platform implementing JWT authentication with email verification, student profile/onboarding management, and a REST API for chat sessions and conversation history.
- Architecture of a provider-agnostic RAG module spanning 2 LLM providers (Mistral primary, Google Gemini fallback), ensuring 100% of answers are grounded in verified documents with precise citations, leveraging PostgreSQL and pgvector for semantic search.
- Delivery of the backend in Python (FastAPI) and SQLAlchemy and the frontend in Next.js/TypeScript, with test-suite automation via pytest.

Environment: *Python, FastAPI, SQLAlchemy, PostgreSQL, pgvector, Mistral, Gemini, Next.js, TypeScript, pytest*

06/2026 - 08/2026 | TechCodex — Ariana, Tunisia

AI Engineer - Computer Vision & Full-Stack Developer

- Fine-tuning and 4-bit quantization of a 4-billion-parameter Vision-Language model (MedGemma 1.5 4B), cutting memory footprint while extracting structured data from handwritten and printed medical prescriptions with optimized local GPU inference.
- Construction of a full-stack architecture connecting a Flask AI backend, a Django REST Framework business API, and a React/Vite frontend for reviewing and validating extraction results.
- Establishment of a human-in-the-loop validation workflow to guarantee data reliability before export.
- Automation of PDF export for validated prescriptions and the full data lifecycle, including anonymization, archiving, and purging.

Environment: *Python, MedGemma, PyTorch, Flask, Django REST Framework, React, Vite, Docker, GPU (CUDA)*

06/2025 - 08/2025 | LabelGarantie — Bizerte, Tunisia

AI Engineer - NLP & Computer Vision

- Training of automatic text-extraction models to process complex documents (PDFs, images) by combining OCR with LLM architectures.
- Design and integration of an end-to-end pipeline covering ingestion, parsing, classification, and Retrieval-Augmented Generation (RAG).
- Industrialization of the complete pipeline through a REST API built with FastAPI.

Environment: *Python, PyTorch, HuggingFace, LangChain, Tesseract, PaddleOCR, FastAPI, REST API*

06/2024 - 08/2024 | LabelGarantie — Bizerte, Tunisia

Front-End Web Developer

- Development of dynamic, responsive UI modules that improved the overall user experience.
- Application of the MVC pattern in Symfony and connection of the front end to REST APIs for seamless data exchange.
- Partnership with the back-end team to debug and resolve API integration issues, with participation in Git-based code reviews.

Environment: *Symfony (PHP), Twig, JavaScript, HTML5, CSS3, MySQL, Git*

PERSONAL PROJECTS

TariqMap - Offline-First Intelligent Road Inspection | Sep 2026 - Present

Offline-first mobile app for road inspection and mapping: on-device detection (YOLOv8, TFLite) of potholes and cracks, geolocation, resilient sync (SQLite, Supabase) and an interactive anomaly map. Built with Flutter, FastAPI and Supabase; currently being improved. Link: github.com/Road-damage-project

Portail Étudiant Intelligent (FSB) | Jan 2026 - May 2026

Full-stack university portal with secure grade management, timetables, signed PDF documents, and a news feed. Integrated a Gemini AI assistant for academic guidance and personalized performance analysis. Built with TypeScript, React, Vite, Tailwind CSS, and Supabase; deployed on Vercel. Link: github.com/AymenJeddou/FSBridge

JobJet - Professional Social Network Platform | Dec 2025 - Jan 2026

Full-stack professional networking application with profile management, job postings, applications, real-time messaging, and notifications, plus automatic PDF CV generation and admin/moderation modules. Built with Node.js, Express.js, MongoDB, and Docker. Link: github.com/AymenJeddou/JobJet

OliveGrove IoT - Smart Farm Monitoring | Sep 2025 - Nov 2025

End-to-end IoT solution for real-time monitoring of environmental parameters (temperature, humidity) in an olive grove, with automated sensor data collection and a web-based visualization dashboard. Link: github.com/AbderrahimNeji/OliveGrove_IOT

TECHNICAL SKILLS

AI & Data: RAG pipelines, LLM integration (Mistral, Gemini, MedGemma), Computer Vision, NLP

Backend & Web: Python, FastAPI, Django REST Framework, TypeScript, React/Next.js

Data & DevOps: PostgreSQL (pgvector), MongoDB, Docker, Git

EDUCATION

09/2024 - Present | Faculté des Sciences de Bizerte — Engineering Degree, Data Specialty (Cycle d'Ingénieur)

09/2022 - 05/2024 | Faculté des Sciences de Bizerte — Integrated Preparatory Cycle - Mathematics, Physics, Computer Science

CERTIFICATIONS

Associate AI Engineer for Data Scientists - DataCamp (Project-Based) · Associate AI Engineer for Developers - DataCamp (Work-Based) · Associate Data Scientist - DataCamp (Project-Based) · Associate Data Engineer - DataCamp (Project-Based) · Associate Data Analyst - DataCamp (Project-Based)

LANGUAGES

Arabic: Native · French: Professional Proficiency (B2) · English: Professional Proficiency (B2)